



SSD20X MultiPanel



© 2020 SigmaStar Technology Corp. All rights reserved.

SigmaStar Technology makes no representations or warranties including, for example but not limited to, warranties of merchantability, fitness for a particular purpose, non-infringement of any intellectual property right or the accuracy or completeness of this document, and reserves the right to make changes without further notice to any products herein to improve reliability, function or design. No responsibility is assumed by SigmaStar Technology arising out of the application or use of any product or circuit described herein; neither does it convey any license under its patent rights, nor the rights of others.

SigmaStar is a trademark of SigmaStar Technology Corp. Other trademarks or names herein are only for identification purposes only and owned by their respective owners.



{SigmaStar Display}

{Smart Display}

{SSD201 + 1.0}

REVISION HISTORY

Revision No.	Description	Date
0.1	Initial release	01/20/2020



1. 概述

Multipanel 主要是将屏参通过 ini 的方式存到板子对应的 flash 分区, 在 panel init 的时候直接去 load 对应的 ini 文件。将 load 屏参的工作放到 sdk 来做, 实现同一个应用程序可以兼容多个 panel 的功能。

2. 使用方法

2.1. 将需要使用的屏参打包进 image

参考 project\board\ini\LCM 下面屏参 ini 的格式，将需要的屏参拷贝至这个目录。

```
[LCD]
m_pPanelName = "SAT070CP50_1024x600";
m_bPanelDither = 0;
m_ePanelLinkType = 0; #0:t11:1:lvds 11:mipi

m_bPanelInvDCLK = 0;
m_bPanelInvDE = 0;
m_bPanelInvHSync = 0;
m_bPanelInvVSync = 0;

m_wPanelHSyncWidth = 48;
m_wPanelHSyncBackPorch = 46;
m_wPanelVSyncWidth = 4;
m_wPanelVBackPorch = 23;
m_wPanelHStart = 98;
m_wPanelVStart = 27;
m_wPanelWidth = 1024;
m_wPanelHeight = 600;

m_wPanelHTotal = 1344;
m_wPanelVTotal = 635;
m_wPanelDCLK = 51;
```

2.2. 指定默认使用的屏参

目前默认屏参通过 project\board\i2m\SSC011A-S01A\config\model\LCM.ini(fastboot 使用 LCM_fastboot.ini) 来指定：

```
[LCM]
PANEL_NAME = "/config/LCM/SAT070AT50H18BH_1024x600_TTL.ini"
CUSTOMER_PANEL_INDEX_GM8775C_1920x1080_LVDS = "/config/LCM/GM8775C_1920x1080_LVDS.ini"; #mipi to lvds panel
CUSTOMER_PANEL_INDEX_RM68200_720x1280_MIPI = "/config/LCM/RM68200_720x1280_MIPI.ini";
CUSTOMER_PANEL_INDEX_SAT070AT50H18BH_1024x600_TTL = "/config/LCM/SAT070AT50H18BH_1024x600_TTL.ini";
CUSTOMER_PANEL_INDEX_SAT070JHH_1024x600_TTL = "/config/LCM/SAT070JHH_1024x600_TTL.ini";
```

如果要动态切换屏参可以通过更改 LCM.ini 的 PANEL_NAME 来指定对应的屏参，或者通过 export LCD= xxx (eg: CUSTOMER_PANEL_INDEX_RM68200_720x1280_MIPI)来指定。

2.3. App 初始化 panel 步骤

导入 multipanel 的措施后, app 初始化屏参只需要做 panel init 就好了:

```

...if (pstDispPubAttr->eIntfType == E_MI_DISP_INTF_LCD)
...{
#if USE_MIPI
    eLinkType = E_MI_PNL_LINK_MIPI_DSI;
#else
    eLinkType = E_MI_PNL_LINK_TTL;
#endif
...}

...stInputPortAttr.u16SrcWidth = LOCAL_VIDEO_W;
...stInputPortAttr.u16SrcHeight = LOCAL_VIDEO_H;
...stInputPortAttr.stDispWin.u16X = LOCAL_VIDEO_X;
...stInputPortAttr.stDispWin.u16Y = LOCAL_VIDEO_Y;
...stInputPortAttr.stDispWin.u16Width = LOCAL_VIDEO_W;
...stInputPortAttr.stDispWin.u16Height = LOCAL_VIDEO_H;

...//MI_DISP_SetPubAttr(0, pstDispPubAttr);
...MI_DISP_Enable(0);
...MI_DISP_BindVideoLayer(0, 0);
...MI_DISP_EnableVideoLayer(0);

...MI_DISP_SetInputPortAttr(0, 0, &stInputPortAttr);
...MI_DISP_EnableInputPort(0, 0);
...MI_DISP_SetInputPortSyncMode(0, 0, E_MI_DISP_SYNC_MODE_FREE_RUN);

...if (pstDispPubAttr->eIntfType == E_MI_DISP_INTF_LCD)
...{
    printf("MI_PANEL_Init: s\n");
    MI_PANEL_Init(eLinkType);
    printf("MI_PANEL_Init: lle\n");
...}
...return 0;
  
```



{SigmaStar Display}
{Smart Display}
{SSD201 + 1.0}